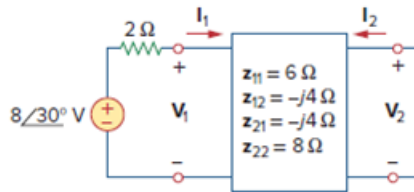


Problem

Calculate I_1 and I_2 in the two-port of Fig. 19.11.

Figure 19.11



Step-by-step solution

Step 1 of 3

Refer to Figure 19.11 in the text book.

Consider the expression for the voltage V_2 in terms of z-parameters.

$$V_2 = z_{21}I_1 + z_{22}I_2$$

Substitute $-j4 \Omega$ for z_{21} , 8Ω for z_{22} and 0 V for V_2 since output port is shorted.

$$0 = (-j4)I_1 + (8)I_2$$

$$(8)I_2 = (j4)I_1$$

$$I_2 = \left(\frac{j4}{8}\right)I_1$$

$$I_2 = (j0.5)I_1$$

$$I_2 = (0.5\angle 90^\circ)I_1 \dots\dots (1)$$

Therefore, the expression for the current I_2 is $(0.5\angle 90^\circ)I_1$.

Step 2 of 3

Consider the expression for the voltage V_1 in terms of the source voltage.

$$V_1 = 8\angle 30^\circ - I_1(2)$$

Consider the expression for the voltage V_1 in terms of z-parameters.

$$V_1 = z_{11}I_1 + z_{12}I_2$$

Substitute $(0.5\angle 90^\circ)I_1$ for I_2 , $8\angle 30^\circ - I_1(2)$ for V_1 , $-j4 \Omega$ for z_{12} and 6Ω for z_{11} .

$$8\angle 30^\circ - (2)I_1 = (6)I_1 + (-j4)(0.5\angle 90^\circ)I_1$$

$$8\angle 30^\circ - (2)I_1 = (6)I_1 + (4\angle -90^\circ)(0.5\angle 90^\circ)I_1$$

$$8\angle 30^\circ - (2)I_1 = (6)I_1 + (2)I_1$$

$$8\angle 30^\circ = (6)I_1 + (2)I_1 + (2)I_1$$

$$8\angle 30^\circ = (10)I_1$$

$$I_1 = \frac{8\angle 30^\circ}{10}$$

$$I_1 = 800\angle 30^\circ \text{ mA}$$

Therefore, the value of the current I_1 is $800\angle 30^\circ \text{ mA}$.

Recall equation (1).

$$\mathbf{I}_2 = (0.5 \angle 90^\circ) \mathbf{I}_1$$

Substitute $800 \angle 30^\circ \text{ mA}$ for $\mathbf{I}_1$.

$$\begin{aligned} \mathbf{I}_2 &= (0.5 \angle 90^\circ)(800 \angle 30^\circ \text{ m}) \\ &= 400 \angle 120^\circ \text{ mA} \end{aligned}$$

Therefore, the value of the current $\mathbf{I}_2$ is $400 \angle 120^\circ \text{ mA}$.